Question # 10 of 10 (Start time: 08:37:31 PM, 19 November 2018)

If for k>0 , d(3,9)= $\frac{1+kd_1(3,9)}{kd_1(3,9)}$ where d_1 represents the usual metric on R then

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Select the correct option

- [x] d(3,9)>1
- [ ] d(3,9)<1
- [ ] d(3,9)=1
- [ ] d(3,9)=0

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Question # 4 of 10 (Start time: 09:15:43 PM, 19 November 2018) Total Marks: 1

If 'k' is a Negative real number, and d is a metric on X then for all x,y belongs to X,

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Select the correct option

Reload Math Equations

●	$kd(x,y) < 0$
○	$kd(x,y) \leqslant 0$
○	$kd(x,y) \geqslant 0$
○	$kd(x,y) > 0$

2.5-5 Theorem (Finite dimension). *If a normed space X has the property that the closed unit ball $M=\{x \mid \|x\| \leqq 1\}$ is compact, then X is finite dimensional.*

Proof. We assume that M is compact but $\dim X=\infty$, and show that this leads to a contradiction. We choose any x_1 of norm 1. This x_1 generates a one dimensional subspace X_1 of X, which is closed (cf. 2.4-3) and is a proper subspace of X since $\dim X=\infty$. By Riesz's lemma there is an $x_2 \in X$ of norm 1 such that

$$\|x_2 - x_1\| \geqq \theta = \frac{1}{2}.$$

The elements x_1, x_2 generate a two dimensional proper closed subspace X_2 of X. By Riesz's lemma there is an x_3 of norm 1 such that for all $x \in X_2$ we have work.

2.6-9 Theorem (Range and null space). *Let T be a linear operator. Then:*

(a) *The range $\mathcal{R}(T)$ is a vector space.*

(b) *If* $\dim \mathcal{D}(T) = n < \infty$, *then* $\dim \mathcal{R}(T) \leqq n$.

(c) *The null space $\mathcal{N}(T)$ is a vector space.*

Long

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Proof. **(a)** We take any $y_1, y_2 \in \mathcal{R}(T)$ and show that $\alpha y_1 + \beta y_2 \in \mathcal{R}(T)$

bijective linear operators, where X, Y, Z are vector spaces (se…21). Then the inverse $(ST)^{-1}\colon Z \longrightarrow X$ *of the product (the compos… ST exists, and*

$$(ST)^{-1} = T^{-1}S^{-1}.$$

Proof. The operator $ST\colon X \longrightarrow Z$ is bijective, so that (ST)… …ts. We thus have

$$ST(ST)^{-1} = I_Z$$

…re I_Z is the identity operator on Z. Applying S^{-1} and usin… …$S = I_Y$ (the identity operator on Y), we obtain in the following basic definition.

2.7-1 Definition (Bounded linear operator). Let X and Y be normed spaces and $T\colon \mathscr{D}(T) \longrightarrow Y$ a linear operator, where $\mathscr{D}(T) \subset X$. The operator T is said to be *bounded* if there is a real number c such that for all $x \in \mathscr{D}(T)$,

Short

$$\|Tx\| \leqq c\|x\|. \tag{1}$$ ∎

In (1) the norm on the left is that on Y, and the norm on the right is that on X. For simplicity we have denoted both norms by the same symbol $\|\cdot\|$, without danger of confusion. Distinction by subscripts ($\|x\|_0$, $\|Tx\|_1$, etc.) seems unnecessary here. Formula (1) shows that a bounded linear operator maps bounded sets in $\mathscr{D}(T)$ onto bounded sets in Y. This motivates the term "bounded operator."

Warning. Note that our present use of the word "bounded" is different from that in calculus, where a bounded function is one whose

Question # 1 of 10 (Start time: 08:44:59 PM, 19 November 2018)

If "d" is a discrete metric on R , then d(3,2)=...

Select the correct option

○	0
●	1
○	2
○	-1

Question # 2 of 10 (Start time: 06:45:51 PM, 19 November 2018)

A sequence (x_n) in a metric space X is said to be a......... if for every $\varepsilon > 0$, there is an N= $N(\varepsilon)$ such that for all m,n > N $d(x_m, x_n) < \varepsilon$

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Select the correct option

- ( ) Geometric Sequence
- ( ) Arithmatic Sequence
- (•) Cauchy sequence
- ( ) Harmonic Sequence

Question # 3 of 10 (Start time: 06:46:42 PM, 19 November 2018)

If Q is dense in R, then $\overline{Q}$ =

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Select the correct option

- R
- Q
- $R\backslash Q$
- ϕ

Question # 4 of 10 (Start time: 06:47:23 PM, 19 November 2018)

The set of all limit points of a set A is called ....of A

Select the correct option

- (●) Derived Set
- ( ) Closure
- ( ) Subset
- ( ) Superset

Question # 5 of 10 (Start time: 06:48:46 PM, 19 November 2018)

The property $d(x, y) \geqslant 0$ of a metric space is called

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○	Reflexive property
●	Non negativity
○	Symmetric property
○	Triangle property

Question # 6 of 10 (Start time: 06:49:52 PM, 19 November 2018)

$C\,[a, b]$ shows the space of all ....defined on [a,b]

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○	Bounded functions
●	Continuous functions
○	Convergent sequences
○	Complex valued functions

Question # 7 of 10 (Start time: 06:51:17 PM, 19 November 2018)

For a metric space (X,d) for all x,y belongs to X $d(x,y) = d(y,x)$ is called ....property

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○	Non negativity
○	Reflexive
●	Symmetric
○	Triangle

Question # 8 of 10 (Start time: 06:52:38 PM, 19 November 2018)

The inverse image of .... set under a continuous mapping is open.

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Select the correct option

- an open
- a closed
- finite
- semi open

Question # 9 of 10 (Start time: 06:53:40 PM, 19 November 2018)

l^p

space is called............ for p=2

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Select the correct option

○	Norm space
○	Metric space
●	Hilbert sequence space
○	Inner product space

Question # 10 of 10 (Start time: 06:54:09 PM, 19 November 2018)

If τ is typology on non-empty set X, then .... intersection of member of belong to τ

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Select the correct option

- infinite
- finite
- arbitrary
- none of these

MTH 641. Paper

Show that absolutely convergent series is normed space,
define complete metric space and....
define linear operator
Prove closure of M is......
X and Y are spaces then T:D(T)- - ->Y is linear operator if T^-1 exists if and only if Tx=0 implies that x=0
Prove Real line is a complete metric spac

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Forwarded

MTH 641. Paper

1): Show that C is a Separable space,
2): compact space,
3): Give two examples of linear operator
4): Prove x belong to closure of M.
5): X and Y are spaces then T:D(T)- - ->Y is linear operator if T^-1 exists.
Show that it is a linear operator. 6
6): Finite dimension prove

7:34 PM

1.1-6 Sequence space l^∞. This example and the next one give a first impression of how surprisingly general the concept of a metric space is.

As a set X we take the set of all bounded sequences of complex numbers; that is, every element of X is a complex sequence

$$x = (\xi_1, \xi_2, \cdots) \qquad \text{briefly}$$

such that for all $j = 1, 2, \cdots$ we have

$$|\xi_j| \leqq c_x$$

where c_x is a real number which may depend on x, but does not depend on j. We choose the metric defined by

$$(9) \qquad d(x, y) = \sup_{j \in \mathbf{N}} |\xi_j - \eta_j|$$

space $X = (X, d)$.

1.3-1 Definition (Ball and sphere). Given a point $x_0 \in X$ and a real number $r > 0$, we define[6] three types of sets:

$$(a) \qquad B(x_0; r) = \{x \in X \mid d(x, x_0) < r\} \qquad \textbf{(Open ball)}$$

$$(1) \quad (b) \qquad \tilde{B}(x_0; r) = \{x \in X \mid d(x, x_0) \leqq r\} \qquad \textbf{(Closed ball)}$$

$$(c) \qquad S(x_0; r) = \{x \in X \mid d(x, x_0) = r\} \qquad \textbf{(Sphere)}$$

In all three cases, x_0 is called the *center* and r the *radius*. ∎

We see that an open ball of radius r is the set of all points in X whose distance from the center of th

the definition immediately implies t

(2) $\qquad S(x_0; r) = \tilde{B}(x_0$

Warning. In working with metric spaces, it is a great advantage that we use a terminology which is analogous to that of Euclidean

...oof of Theorem 1.3-4

...age N_0 of N is open, since N ... so contains a δ-neighborhood of x_0, which is mapped into N because N_0 is mapped into N. Consequently, by the definition, T is continuous at x_0. Since $x_0 \in X$ was arbitrary, T is continuous. ∎

We shall now introduce two more concepts, which are related. Let M be a subset of a metric space X. Then a point x_0 of X (which may or may not be a point of M) is called an **accumulation point** *of M* (or *limit point of M*) if every neighborhood of x_0 contains at least one point $y \in M$ distinct from x_0. The set consisting of the points of M and the accumulation points of M is called the **closure** *of M* and is denoted by

$$\bar{M}.$$

and only if X is countable. (Cf. 1.1-8.)

Proof. The kind of metric implies that no proper subset of X can be dense in X. Hence the only dense set in X is X itself, and the statement follows.

1.3-9 Space l^∞. *The space l^∞ is not separable.* (Cf. 1.1-6.)

Proof. Let $y = (\eta_1, \eta_2, \eta_3, \cdots)$ be a sequence of zeros and ones. Then $y \in l^\infty$. With y we associate the real number $\hat{y}$ whose binary representation is

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in the interval [0, 1] is

1.4-2 Lemma (Boundedness, limit). *Let $X=(X, d)$ be a metric space. Then:*

(a) *A convergent sequence in X is bounded and its limit is un[illegible]ue.*

(b) *If $x_n \longrightarrow x$ and $y_n \longrightarrow y$ in X, the[illegible] $d(x_n, y_n) \longrightarrow d(x, y)$.*

Proof. **(a)** Suppose that $x_n \longrightarrow x$. [illegible]hen, [illegible]ing [illegible] we can find an N such that $d(x_n, x)<1$ for all $n>N$. Hence by the triangle inequality (M4), Sec. 1.1, for all n we have $d(x_n, x)<1+a$ where

...strates that the concept of ... he sequence itself but also ... ce lies. In other words, a ... on its own" but it must

... cient for convergen..., it is worth noting that it continues to be necessary for convergence. In fact, we readily obtain the following result.

1.4-5 Theorem (Convergent sequence). *Every convergent sequence in a metric space is a Cauchy sequence.*

Proof. If $x_n \longrightarrow x$, then for every $\varepsilon > 0$ there is ... N ... such that mapping.

1.4-8 Theorem (Continuous mapping). *A mapping $T\colon X \longrightarrow Y$ of a metric space (X, d) into a metric space $(Y, \tilde{d})$ is continuous at a point*

$x_0 \in X$ *if and only if*

$$x_n \longrightarrow x_0 \qquad \text{i}$$

Proof. Assume T to be c

various other applications.

1.6-1 Definition (Isometric mapping, isometric spaces). Let $X = (X, d)$ and $\tilde{X} = (\tilde{X}, \tilde{d})$ be metric spaces. Then:

(a) A mapping T of X into $\tilde{X}$ is said to be *isometric* or an *isometry* if T preserves distances, that is, if for all $x, y \in X$,

$$\tilde{d}(Tx, Ty) = d(x, y),$$

where Tx and Ty are the images of x and y, respectively.

SHORT

(b) The space X is said to be *isometric* with the space $\tilde{X}$ if there exists a bijective[9] isometry of X onto $\tilde{X}$. The spaces X and $\tilde{X}$ are then

(b) The space X is said to be *isometric* with the space $\tilde{X}$ if there exists a bijective[9] isometry of X onto $\tilde{X}$. The spaces X and $\tilde{X}$ are then called *isometric spaces.* ∎

Hence isometric spaces may differ at most by the nature of their points but are indistinguishable from the viewpoint of metric. And in any study in which the nature of the points does not matter, we may regard the two spaces as identical—as two copies of the same "abstract" space.

We can now state and prove the theorem that every metric space can be completed. The space $\hat{X}$ occurring in this theorem is called the **completion** of the given space X.

1.6-2 Theorem (Completion). *For a metric space $X=(X,d)$ there exists a complete metric space $\hat{X}=(\hat{X},\hat{d})$ which has a subspace W that is isometric with X and is dense in $\hat{X}$. This space $\hat{X}$ is unique except for isometries, that is, if $\tilde{X}$ is any complete metric space having a dense subspace $\tilde{W}$ isometric with X, then $\tilde{X}$ and $\hat{X}$ are isometric.*

[9] One-to-one and onto. For a review of some elementary concepts related to mappings, see A1.2 in Appendix 1. Note that an isometric mapping is always injective. (Why?)

Proof. The proof is somewhat lengthy but straightforward. We subdivide it into four steps (a) to (d). We construct:

(a) $\hat{X}=(\hat{X},\hat{d})$

(b) an isometry T of X onto W, where $\bar{W}=\hat{X}$.

Then we prove:

(c) completeness of $\hat{X}$,

(d) uniqueness of $\hat{X}$, except for isometries.

In analysis, infinite dimensional vector spaces are of greater interest than finite dimensional ones. For instance, $C[a, b]$ and l^2 are infinite dimensional, whereas $\mathbf{R}^n$ and $\mathbf{C}^n$ are n-dimensional.

If $\dim X = n$, a linearly independent n-tuple of vectors of X is called a **basis** for X (or a *basis in* X). If $\{e_1, \cdots, e_n\}$ is a basis for X, every $x \in X$ has a unique representation as a linear combination of the basis vectors:

$$\alpha_1 e_1 + \cdots + \alpha_n e_n.$$

For instance, a basis for $\mathbf{R}^n$ is

$$\|y_m - y\| = \left\| \sum_{j=1}^{n} (\alpha_j^{(m)} - \alpha_j) e_j \right\| \leq \sum_{j=1}^{n} |\alpha_j^{(m)} - \alpha_j| \, \|e_j\|.$$

On the right, $\alpha_j^{(m)} \longrightarrow \alpha_j$. Hence $\|y_m - y\| \longrightarrow 0$, that is, $y_m \longrightarrow y$. This shows that (y_m) is convergent in Y. Since (y_m) was an arbitrary Cauchy sequence in Y, this proves that Y is complete. ∎

From this theorem and Theorem 1.4-7 we have

2.4-3 Theorem (Closedness). *Every finite dimensional subspace Y of a normed space X is closed in X.*

We shall need this theorem at several occasions in our further work.

Note that infinite dimensional subspaces need not be closed.

2.5-5 Theorem (Finite dimension). *If a normed space X has the property that the closed unit ball $M = \{x \mid \|x\| \leqq 1\}$ is compact, then X is finite dimensional.*

Proof. We assume that M is compact but $\dim X = \infty$, and show that this leads to a contradiction. We choose any x_1 of norm 1. This x_1 generates a one dimensional subspace X_1 of X, which is closed (cf. 2.4-3) and is a proper subspace of X since $\dim X = \infty$. By Riesz's lemma there is an $x_2 \in X$ of norm 1 such that

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Long

(a) *The range $\mathscr{R}(T)$ is a vector space.*

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(c) *The null space $\mathscr{N}(T)$ is a vector space.*

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Proof. **(a)** We take any $y_1, y_2 \in \mathscr{R}(T)$ and show that $\alpha y_1 + \beta y_2 \in \mathscr{R}(T)$